THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Sample Test 2

Please write down your name and university number. Precise and adequate explanations should be given to each problem.

1. Define $f : X \to \mathbb{R}$ by $f(x) = x^2 \sin\left(x^3 + \dfrac{1}{x}\right) - 1$ where $X = \mathbb{R} - \{0\} = \{x \in \mathbb{R} : x \neq 0\}$.

   (a) (2 points) Prove that 0 is a limit point of X by definition.

   (b) (5 points) Using the ε-δ definition, compute $\lim\limits_{x \to 0} f(x)$.

2. Let $\{a_n\}$ be a sequence of positive real numbers such that the series $\sum\limits_{n=1}^{\infty} a_n$ converges.

   (a) (5 points) Prove that $\sum\limits_{n=1}^{\infty} \dfrac{a_n}{a_{n+1}}$ diverges.

   (b) (2 points) Give an example for which $\sum\limits_{n=1}^{\infty} \dfrac{a_{n+1}}{a_n}$ converges. (Note that you should briefly explain why the series converges.)

3. (7 points) Let $\{b_n\}$ be a sequence of real numbers such that $b_1 = 2$ and $b_{n+1} = \dfrac{2b_n^2 + b_n + 1}{b_n - 1}$ for any positive integer n. Prove that $\sum\limits_{n=1}^{\infty} \dfrac{1}{b_n}$ converges and find this sum. (Note that 4 points are given for proving the convergence without finding the sum.)

4. (7 points) Suppose $f : [0,1] \to \mathbb{R}$ is a continuous function. Define a function $g : [0,1] \to \mathbb{R}$ by $g(x) = \sup\{f(y) : 0 \leqslant y \leqslant x\}$. Prove that g is continuous at every $c \in (0,1)$.

5. Prove or disprove each of the following. State clearly whether it is true or false.

   (a) (4 points) Let $\{a_n\}$ be a sequence of real numbers. If the series $\sum\limits_{n=1}^{\infty} a_n$ converges, then the series $\sum\limits_{n=1}^{\infty} \left(a_n + (-1)^{n+1}\right)$ converges.

   (b) (4 points) Let X be a subset of $\mathbb{R}$ and let a be a limit point of X. Then for any $\delta > 0$, the intersection of X and the interval $(a - \delta, a + \delta)$ contains infinitely many real numbers.

   (c) (4 points) Let $f : \mathbb{R} \to \mathbb{R}$ be a function. If the composition $f \circ f$ is uniformly continuous, then f must be continuous.